

D2: Record Duration and Welch Segment Length

8 October 2026 (rerun)

How long must we record, and which L should we use?

Amir Dollie (DLLAMI004), EEE4022S

SHARC buoy project Code: `runD2RecordDuration.m`, commit `e9eacc7` Results: `Results/D2/D2_20261008.173956` (196 s)

Rerun after the amplitude amendment (8 Oct). The EMM FRFs used by the 2 Oct run were low by $\alpha(\omega) = H\omega^2/g$ (1.7 to 13.85 over the band): the solver's incident wave has amplitude $1/\alpha$, not 1. `computeSensorFRF` now multiplies by α and the twin cache was migrated exactly (`cfg.meta.changes`, 2026-10-08). All settings are unchanged. This summary supersedes D2_20261002.114914. The corrected signal level raises the SNR, so σ falls about $8\times$; the Welch bias and F both change, and d_W rises.

Summary

Question. For each Welch segment length L , how does record duration trade random error against finite-resolution (Welch) bias, and what is the shortest record that meets the pre-registered Identifiable criteria? **Outcome (LSM6DSV16X).** Precision is no longer the constraint: $\sigma_{\ln \beta} \leq 1\%$ is met by the shortest admissible record ($K \geq 8$) at every L . The shortest Identifiable record is **7.7 min with $L = 1024$** ; with $L = 2048$ the window is **[15.4, 143.3] min**, so the 109 min production record is Identifiable ($\sigma_{\ln \beta} = 0.050\%$, $d_W = 0.436$) but now sits in the upper quarter of its window. **One regime.** Every sensor is bias-limited: the upper window edge, set by d_W , decides. The datasheet-grade case is Not identifiable at 109 min with $L = 2048$ ($d_W = 1.21$) and needs $L = 4096$. **Recommendation.** Keep $L = 2048$, 109 min as the frozen baseline (the setting every study uses), and report $L = 4096$ as the preferred setting for long records once it is validated (E1 gate extension, M1 spot check).

1 What the study does

The floe (p_0), layout (Level 2C, reference at $0.9R$), sea state (JONSWAP $H_s = 0.05$ m, $\omega_p = 5$ rad/s) and EMM field are fixed; only the record length T and the Welch segment length L change. For each of the 6 noise cases, 7 durations (10 to 240 min, $\Delta t = 0.1$ s, so $N = 600T$) and 4 segment lengths, the E1 engine gives $\sigma_{\ln \beta}$, $\sigma_{\ln R}$, the Welch-bias significance d_W , $\kappa(F)$ and the class. No new EMM solves are needed. The 109 min point uses the production record ($N = 65536$) and reproduces the corrected E1 headline exactly ($\sigma_{\ln \beta} = 0.0499\%$, $d_W = 0.436$, `validateAlphaMigration`).

Admissibility. With 50% overlap, a record of N samples gives $K \approx 2N/L - 1$ segments. A combination is admitted only if $K \geq 8$ (pre-registered); 24 of the 28 combinations are admitted:

$K \setminus T$ (min)	10	20	30	60	109	180	240
$L = 512$		22	45	69	139	255	420
$L = 1024$		10	22	34	69	127	209
$L = 2048$		4	10	16	34	63	104
$L = 4096$		1	4	7	16	31	51

Grey: $K < 8$, inadmissible, not evaluated.

Why a record can be too long. At fixed L the random error falls with the effective number of averages, while the Welch bias of the transmissibility estimate does not depend on N :

$$\sigma \propto \frac{1}{\sqrt{N_{\text{eff}} - 1}}, \quad \delta\theta_W = \text{const.} \quad \implies \quad d_W = \sqrt{\delta\theta_W^T F \delta\theta_W} \propto \sqrt{N_{\text{eff}} - 1}.$$

The same bias therefore becomes *more* significant as the record grows. For each L the Identifiable region ($s_{\max} \leq 1\%$, $d_W \leq 0.5$, $\kappa \leq 10^8$) is a window $[T_{\text{lo}}, T_{\text{hi}}]$: too short and σ (or $K \geq 8$) fails, too long and d_W fails. Both scalings hold exactly on the grid (deviation 0.00%), so window edges are found on a 0.1 min grid. The preferred L is the one Identifiable earliest; ties go to the widest window. A secondary measure, $\text{RMSE} = \sqrt{\sigma_{\ln \beta}^2 + \delta \ln \beta_W^2}$, combines both errors.

2 Results

Headline: shortest Identifiable record and the production setting (109 min, $L = 2048$).

Noise case	$\sigma_{\ln \beta}$	d_W	Shortest Identifiable	Preferred L	Production verdict
Datasheet grade	0.020%	1.209	15.4 min	2048	Not identifiable ($d_W > 0.5$); use $L = 4096$
Three IMUs fused	0.043%	0.487	7.7 min	1024	Identifiable; 14.2 $\times$ minimum
LSM6DSV16X	0.050%	0.436	7.7 min	1024	Identifiable; 14.2$\times$ minimum
G3 assumption	0.061%	0.391	7.7 min	1024	Identifiable; 14.2 $\times$ minimum
ICM-42688-P	0.063%	0.387	7.7 min	1024	Identifiable; 14.2 $\times$ minimum
LSM6DS3TR-C	0.150%	0.237	5.2 min	512	Identifiable; 21.0 $\times$ minimum

$\sigma_{\ln \beta}$ and d_W at the production setting. Pre-amendment (2 Oct) for comparison, LSM6DSV16X: $\sigma_{\ln \beta} = 0.36\%$, $d_W = 0.309$, shortest 17.1 min.

Identifiable windows $[T_{lo}, T_{hi}]$ in minutes (search to 600 min). Every T_{lo} except LSM6DS3TR-C at $L = 512$ is set by the $K \geq 8$ rule, not by σ .

Noise case	$L = 512$	$L = 1024$	$L = 2048$	$L = 4096$
Datasheet grade	never	never	[15.4, 22.1]	[30.8, 300.3]
Three IMUs fused	never	[7.7, 11.0]	[15.4, 116.0]	[30.8, >600]
LSM6DSV16X	never	[7.7, 15.3]	[15.4, 143.3]	[30.8, >600]
G3 assumption	never	[7.7, 22.1]	[15.4, 177.4]	[30.8, >600]
ICM-42688-P	never	[7.7, 23.0]	[15.4, 180.9]	[30.8, >600]
LSM6DS3TR-C	[5.2, 8.1]	[7.7, 52.0]	[15.4, 474.4]	[30.8, >600]

RMSE-optimal L ($\ln \beta$) by duration: the optimum moves to longer segments as data accumulates.

T (min)	10	20	30	60	109	180	240
All cases except LSM6DS3TR-C	1024	2048	2048	4096	4096	4096	4096
LSM6DS3TR-C	1024	2048	2048	2048	4096	4096	4096

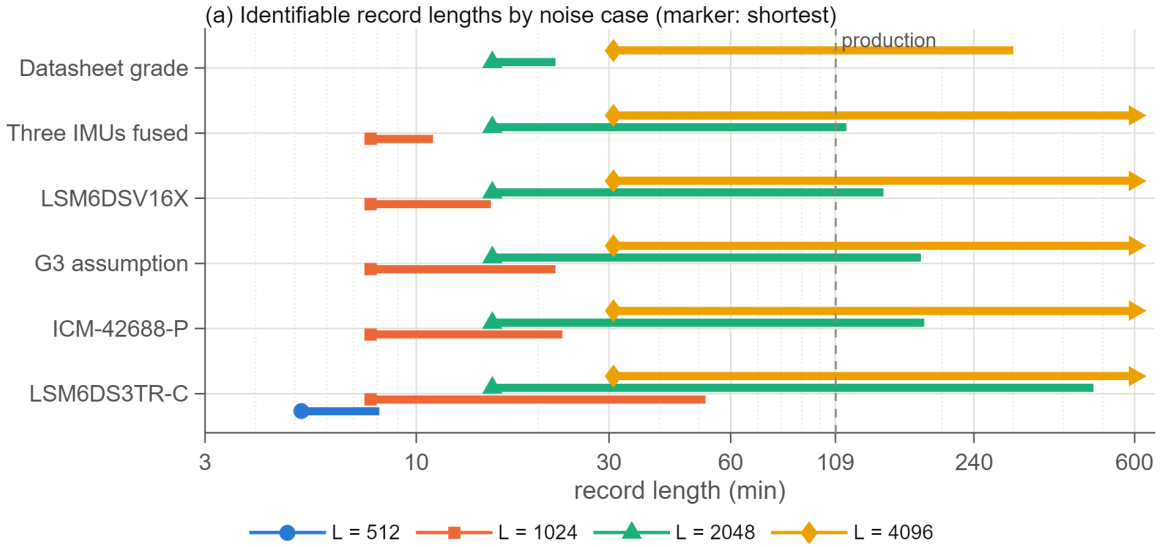


Figure 1: Identifiable record-length windows for each sensor-noise case and Welch segment length L (corrected FRF). Each marker denotes the shortest record satisfying the pre-registered Identifiable criteria ($s_{\max} \leq 1\%$, $d_W \leq 0.5$, $\kappa(F) \leq 10^8$); the horizontal line shows the interval over which they remain satisfied. Arrows indicate windows open beyond the 600 min search limit. The dashed vertical line marks the 109 min production record.

3 Interpretation

1. **The problem is now bias-limited for every sensor.** With the corrected signal level, $\sigma_{\ln \beta} \leq 1\%$ is reached by the shortest admissible record at every L , so the lower window edge is the $K \geq 8$ rule and the upper edge, set by d_W , is what decides. The earlier “noisy sensors need long records” regime has gone: the LSM6DS3TR-C needed 205 min before the amendment and needs 5.2 min now.
2. **$L = 2048$, 109 min remains valid for the LSM6DSV16X, with less margin.** It is Identifiable ($d_W = 0.436$), inside the window [15.4, 143.3] min, but no longer central: 109 min is about three quarters of the way across the window. $L = 1024$ closes at 15.3 min, so $L = 2048$ is still the more robust of the two validated segment lengths.
3. **The bias budget is mostly spent.** At the production setting $d_W = 0.436$ already uses most of the $d_{\text{tot}} \leq 0.5$ allowance before any systematic error (noise-floor error R3, γ error R4, model mismatch V1, directional forcing V2) is added. Those tolerances will be correspondingly tighter at $L = 2048$, 109 min.
4. **$L = 4096$ is the natural setting for long records.** It is RMSE-optimal from 60 min upward for every quiet sensor, its window is open beyond 600 min for all but the datasheet case, and it is the only segment length that makes the datasheet case Identifiable at 109 min. It is not adopted now: it lies outside the E1 validation gate (512 to 2048) and the frozen baseline is retained for comparability across studies. Adopting it needs an E1 gate point at $L = 4096$ and an M1 spot check.
5. **$L = 512$ is ruled out.** Its window smoothing pushes d_W above 0.5 for every case except the noisiest sensor over 5 to 8 min.
6. **The minima are lower bounds, not targets.** Every minimum sits at the $K \geq 8$ boundary, where the covariance model is least tested (E1 was validated at $K = 63$). E1 is a local Fisher prediction (about 5 to 10% optimistic

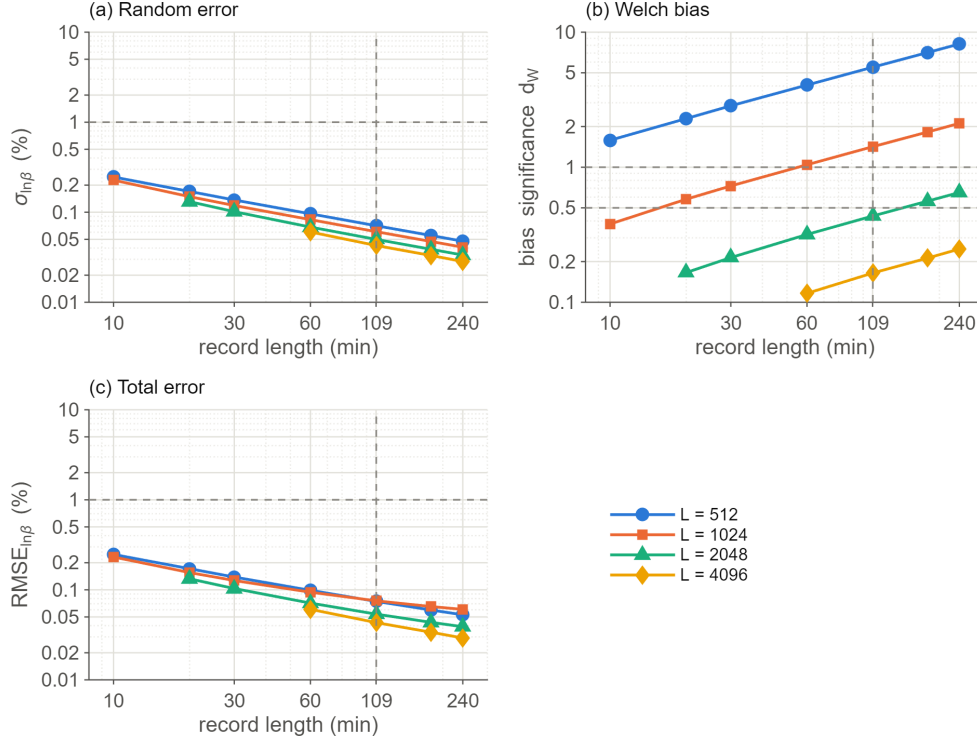


Figure 2: Effect of record duration and Welch segment length on the LSM6DSV16X case (corrected FRF). (a) $\sigma_{\ln \beta}$, 1% threshold dashed. (b) Welch-bias significance d_W , thresholds $d_W = 0.5$ and $d_W = 1$ dashed. (c) $\text{RMSE}_{\ln \beta} = \sqrt{\sigma_{\ln \beta}^2 + \delta \ln \beta_W^2}$. Only combinations with $K \geq 8$ are shown. The dashed vertical line marks the 109 min production record.

against G3 before the amendment; the Monte Carlo references are being regenerated), and other floes (A1), sea states (R2) and real noise (R1) will move these numbers. A field record is also not stationary for hours.

7. **The old bias rule of thumb remains unreliable.** $d \approx 0.26 \sqrt{N/65536} (2048/L)^2$ was fitted to the pre-amendment field; against the corrected engine it gives 0.44 to 1.07 of d_W (G3 noise). Use the engine.

Requirement for the experiment and the thesis

At p_0 with the LSM6DSV16X, $L = 2048$ is Identifiable from 15.4 to 143 min; the 109 min production record is inside this window with $d_W = 0.436$. Keep $L = 2048$, 109 min as the frozen baseline for every study. The design result for the thesis is that the inverse is bias-limited: longer records need longer segments, and $L = 4096$ is the recommended setting for records of an hour or more once validated. Do not use $L = 512$.

Outputs (Results/D2/): D2_20261008.173956.log/.mat/_summary.txt; D2_sweep.csv; D2_min_duration.csv; D2_rmse_optimal.L.csv; figures/D2_fig1.duration_lsm6dsv16x, figures/D2_fig2.identifiable_window (.pdf/.png). Superseded: D2_20261002.114914 (pre-amendment).